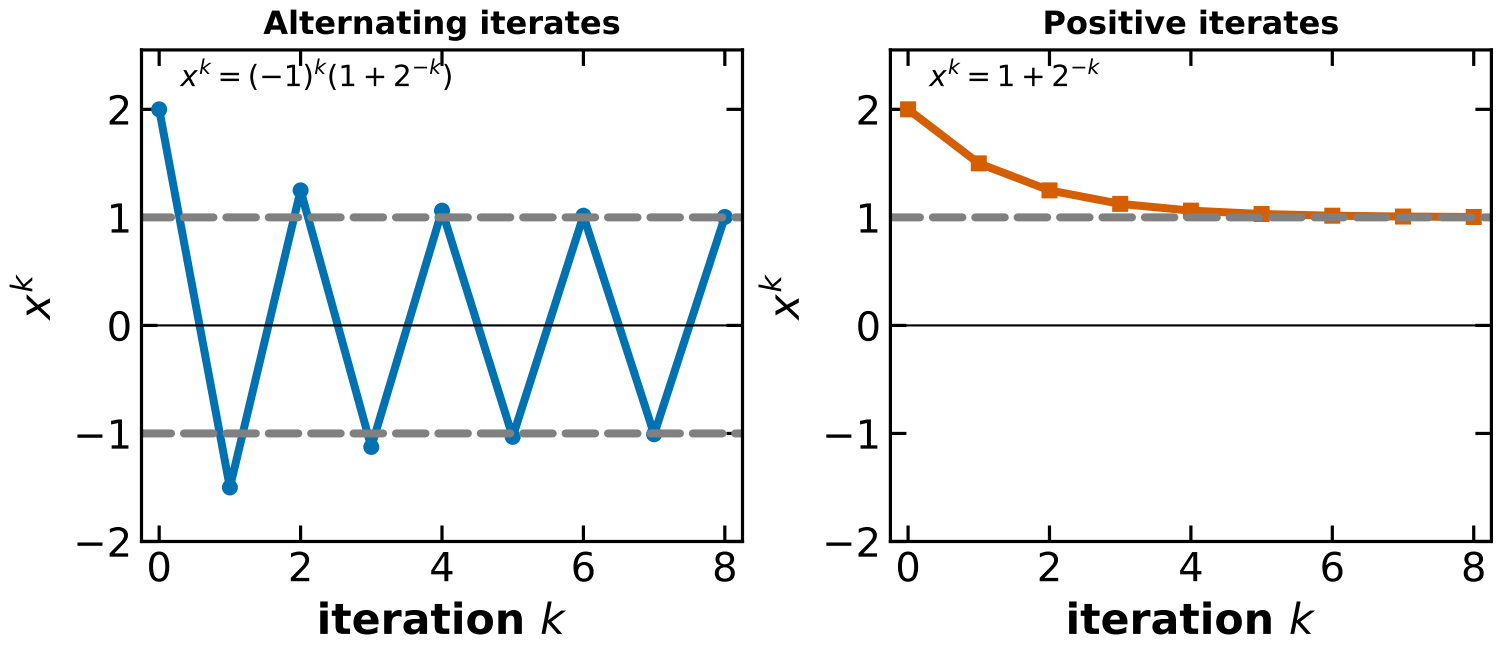
Alternating iterates
$x^k = (-1)^k(1 + 2^{-k})$
x^k
iteration k
Positive iterates
$x^k = 1 + 2^{-k}$
x^k
iteration k